

Md. Mahfujur Rahman

Machine Learning Engineer & AI Researcher

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SUMMARY

Machine Learning Engineer and Researcher specializing in deep learning, computer vision, and end-to-end production AI architectures. Experienced in training, fine-tuning, and deploying convolutional neural networks for oncology histopathology classification and aquatic epidemiology. Competitive programming background as an ICPC Regionalist with strong foundations in algorithms, optimization, and high-performance backend systems.

TECHNICAL EXPERTISE

Machine Learning:	PyTorch, TensorFlow, Keras, OpenCV, CNNs, Transfer Learning, Scikit-learn, ONNX
Languages:	Python, C++, TypeScript, JavaScript, SQL, Bash
Backend & Cloud:	FastAPI, Docker, PostgreSQL, SQLAlchemy, REST APIs, Redis, Vercel, Render
Algorithms:	Data Structures, Graph Algorithms, Dynamic Programming (ICPC Regionalist)

RESEARCH PUBLICATIONS

Deep Convolutional Neural Networks for Automated Cancer Histopathology Classification

Published in Peer-Reviewed Venue | Primary Author & AI Architect

- Developed an end-to-end deep learning pipeline for histopathological tissue image classification with 98.7% test accuracy.
- Implemented data augmentation and transfer learning architectures, significantly reducing false negatives in clinical triage.

Deep Learning Framework for Aquatic Epidemiological Diagnostics and Fish Disease Detection

Research Contribution | Lead Researcher

- Designed an automated computer vision diagnostic tool classifying multiple aquatic disease phenotypes under variable water turbidity.
- Evaluated robustness using confusion matrix analysis and cross-validation across distinct species datasets.

FEATURED ML & SOFTWARE PROJECTS

OncoVision AI — Clinical Cancer Diagnostic & Triage Platform

Tech Stack: PyTorch, FastAPI, React 19, TypeScript, Docker, PostgreSQL, Render, Vercel

- Engineered an asynchronous, containerized microservice serving low-latency deep learning inference.
- Built an interactive React dashboard with progressive skeletons, class probability distributions, and instant prefetching.

Aquatic Epidemiology Classification Engine

Tech Stack: Python, PyTorch, OpenCV, Scikit-learn, FastAPI

- Built dataset preprocessing pipelines handling color normalization, edge feature extraction, and bounding box validation.
- Deployed model inference endpoint with sub-100ms response time for field environmental evaluation.

HONORS & COMPETITIVE PROGRAMMING

- ICPC Regionalist:** Qualified and competed in the prestigious International Collegiate Programming Contest (ICPC).
- Solved 1,000+ algorithmic and mathematical optimization problems across Codeforces, LeetCode, and regional platforms.
- Active technical writer and researcher sharing insights on deep learning architectures and production deployment.